Software Requirements Specification Document (SRS)

Smart Study Planner with AI Suggestions

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Name: P.A.T.Hansamali

Reg.No: DEH/IT/2324/F/0068

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1)Introduction

1.1 Purpose

This document describes the requirements of the Smart Study Planner with AI Suggestions system. It defines system features, functionalities and constraints required for development and implementation.

1.2 Scope

The system is a web-based application that helps students plan their studies efficiently. It allows users to input subjects, deadlines, difficulty levels and available study time and generates a personalized study timetable with AI-based suggestions.

Ex:- If a student has an exam in 2 days and another in 1 week, the system will prioritize the nearer exam and allocate more study time to it.

1.3 Definitions, Acronyms

- ✓ AI – Artificial Intelligence
- ✓ DB – Database
- ✓ UI – User Interface
- ✓ API – Application Programming Interface

2) Overall Description

2.1 Product Perspective

This system is a standalone web application designed to assist students in managing their study schedules. It interacts with a database to store and process user data.

2.2 Product Functions

- ✓ User registration and login
- ✓ Subject and task management
- ✓ Timetable generation
- ✓ AI-based study suggestions
- ✓ Progress tracking

These functions work together to help users organize their studies and improve productivity.

2.3 User Characteristics

- ✓ Students with basic computer knowledge

2.4 Assumptions and Dependencies

- ✓ Users have internet access
- ✓ System requires a working database
- ✓ Users provide accurate inputs such as deadlines and difficulty levels

3) External Interface Requirements

3.1 User Interfaces

- ✓ Login page
- ✓ Dashboard
- ✓ Task management screen
- ✓ Timetable view
- ✓ Progress tracking screen
- ✓ Subject page
- ✓ AI suggestions page
- ✓ Calander page
- ✓ Settings page

3.2 Hardware Interfaces

PC, Laptop or Mobile device

3.3 Software Interfaces

Database: MongoDB

Backend API: Node.js

4) Functional Requirements

FR1: User Registration

The system shall allow users to create an account.

FR2: User Login

The system shall allow users to log in using a username and password.

FR3: Manage Subjects

The system shall allow users to add, edit and delete subjects.

FR4: Manage Tasks

The system shall allow users to add, edit and delete study tasks.

FR5: Timetable Generation

The system shall generate a study timetable based on user input such as subjects, deadlines, difficulty levels and available time.

Ex:- If a subject has a near deadline, it will be scheduled earlier in the timetable.

FR6: AI Suggestions

The system shall analyze deadlines and difficulty levels using rule-based decision logic to provide study recommendations.

Ex:- If a subject is marked as difficult or has a near deadline, the system allocates more study time and prioritizes it.

FR7: Progress Tracking

The system shall display user progress using charts or percentage completion.

FR8: Notifications

The system shall notify users about upcoming deadlines and pending tasks.

FR9: Difficulty Level Input

The system shall allow users to mark subjects as Easy, Medium or Hard.

5) Non-Functional Requirements

5.1 Performance

The system should respond within 2 seconds.

5.2 Security

User passwords must be encrypted

Only authorized users can access their data

5.3 Usability

The system should be user-friendly and easy to navigate for users with basic computer knowledge.

5.4 Reliability

The system should be available with minimal downtime.

6) System Requirements

6.1 Hardware Requirements

Laptop

Mobile device

6.2 Software Requirements

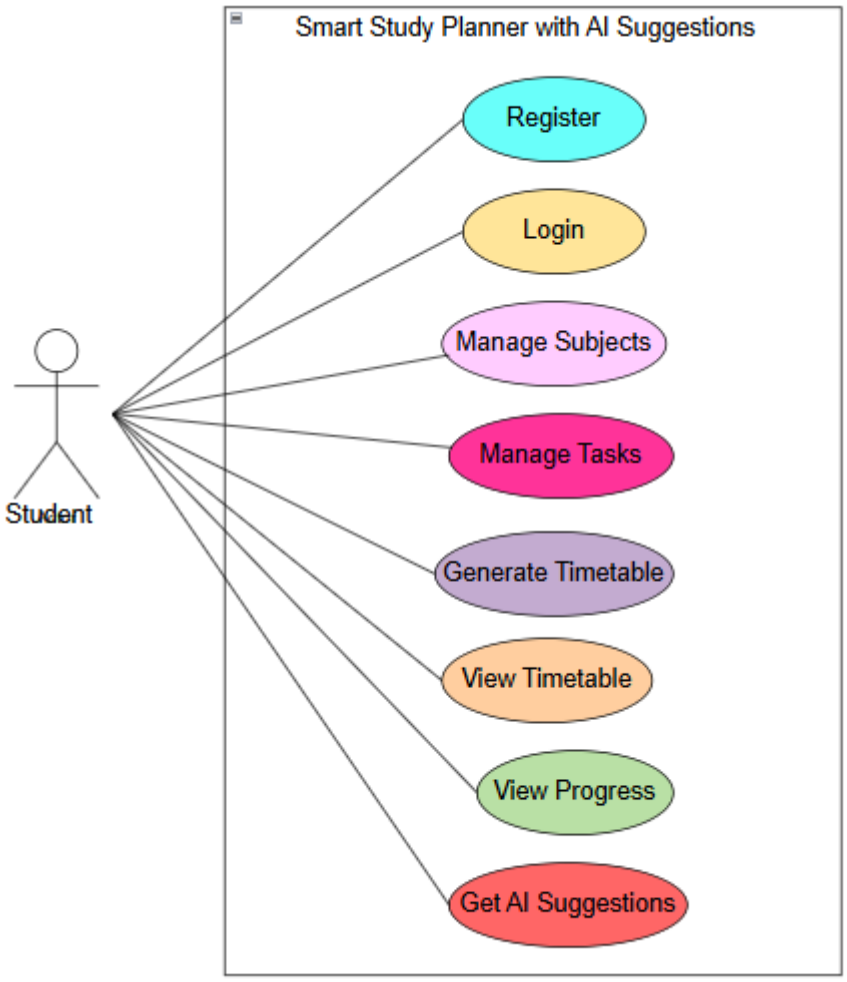
Operating System: Windows

Web Browser: Chrome / Edge

6.3 Tools & Technologies

- ✓ Frontend: React.js
- ✓ Backend: Node.js
- ✓ Database: MongoDB
- ✓ Tools: Visual Studio Code

7) Use Case Diagram (UML)



1) Use Case Descriptions

- **Use Case 1: User Registration**

  Actor: Student

  Description: User creates a new account

  Precondition: User is not registered

  Main Flow:

  - User enters details

- System validates data
- System saves user in database

Outcome: Account created

- **Use Case 2: User Login**

  Actor: Student

  Description: User logs into the system

  Precondition: User already registered

  Main Flow:

  - User enters email and password
  - System verifies credentials
  - Access granted

  Outcome: Dashboard displayed

- **Use Case 3: Generate Timetable**

  Actor: Student

  Description: System generates study timetable

  Precondition: Subjects and tasks added

  Main Flow:

  - User clicks generate
  - System analyzes deadlines and difficulty
  - Timetable created

  Outcome: Timetable displayed

- **Use Case 4: Manage Tasks**

  Actor: Student

  Description: User adds or updates tasks

  Precondition: User logged in

  Main Flow:

  - User adds task
  - System stores task

  Outcome: Task list updated

Other Requirements

❖ Constraints

- ✓ Internet connection required
- ✓ System limited to rule-based AI
- ✓ User must provide correct input data
- ✓ Web-based only

❖ Future Enhancements

- ✓ Mobile application
- ✓ Machine Learning based AI
- ✓ Email notifications
- ✓ LMS integration
- ✓ Advanced analytics dashboard